

Capstone Project Proposal (Data Analysis Pathway)

Project Title: Electrical Outages and Economic Indicators in U.S. Counties, 2022-2024

Proposed By: Jesse Garrido

GitHub Repository: <https://github.com/jessegarrido/usa-electrical-outages>

Project Overview

This project will present correlations between economic indicators and instances of electrical outages across US counties in recent history. The investigation will aim to show the relative extent to which wealth may insulate users from loss of electrical service.

Data Sources

- **Event-correlated Outage Dataset in America**, April 25, 2025
 - Source: Pacific Northwest Laboratory, U.S. Department of Energy
 - Relevant Fields: event_id, state_event, Event Type, fips, state, county, start_time, end_time, duration, min_customers, max_customers, mean_customers
 - Link: <https://catalog.data.gov/dataset/event-correlated-outage-dataset-in-america>
- **GDP and Personal Income Summary by County**, 2022-2024

- **Public Assistance Funded Project Summaries - v1**

Research Objectives

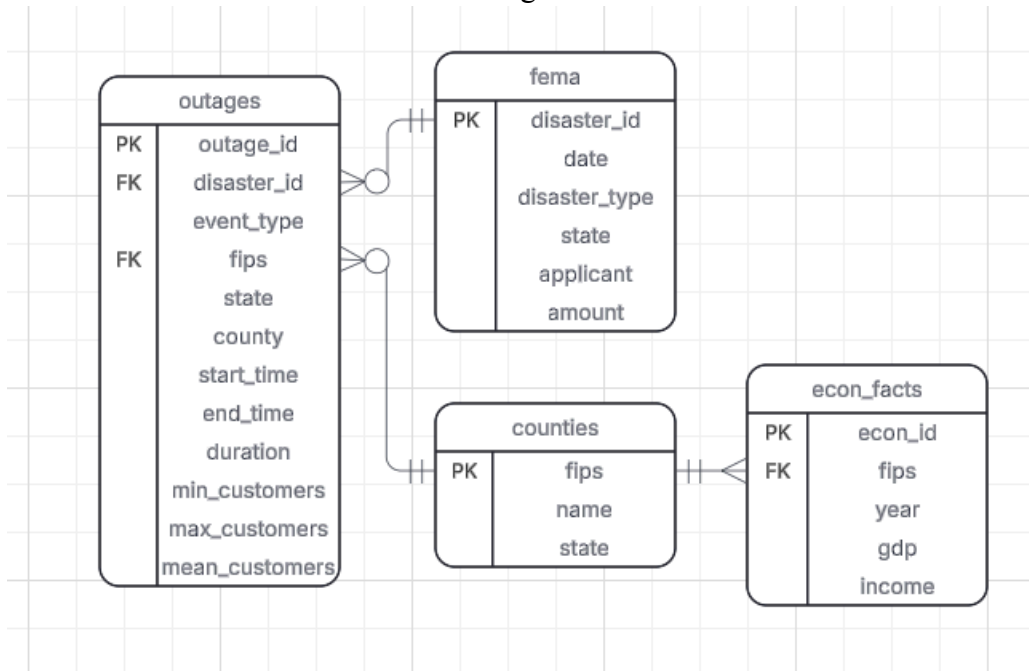
Primary Question: To what extent does wealth insulate users in the U.S. from instances and duration of electrical outages?

Secondary/Exploratory Questions: Are trends observed in intensity and causes of electrical outages? What FEMA spending was associated with specific outage events?

As weather disasters increase in duration and intensity, this research investigates trends in impacts to electrical service, and whether disparity of income correlates to disparity of impact.

Data Preparation Approach

- Electrical outage dataset aligns to economic data on FIPS code. The outage dataset will align to FEMA disaster funding on a unique id assigned to each disaster, after matching ids to weather disasters based on date and location.
- Economic data was joined into one set that includes both gdp and income values. Electrical outage data will be joined into one set from files separated by year.
- New columns will be created to re-categorize event types, align matching disaster ids, and reformat the econ_facts table
- The database will be structured according to the ERD below



Current Status

- To date, project datasets have been identified and inspected.

- A function was developed to parse motivating event descriptions into four categories (instead of thirty-seven)
- Count, duration, and customer-hours of outage events were normalized to evaluate relative distribution across U.S. states.
- Early plots compare the motivating events and locations of electrical outages in just one year.
- Economic and electrical outage data is shown to be compatible by python statement that shows 2023 electrical outages joined on FIPS to economic data for Jefferson County, KY.

```

1 merged_df = df.merge(econ_df, on='fips')
2 merged_df[['fips', 'event_id', 'Event Type', 'county', 'income2023', 'gdp2023']][merged_df['fips'] == 21111]

```

int64	fips int64	event_id object	Event Type object	county object	income2023 object	gdp2023 object
8,199	21,111	Kentucky-3	Vandalism	Jefferson	53172643	62314592
8,224	21,111	Kentucky-4	System Operations	Jefferson	53172643	62314592
8,225	21,111	Kentucky-4	System Operations	Jefferson	53172643	62314592
8,226	21,111	Kentucky-4	System Operations	Jefferson	53172643	62314592
8,248	21,111	Kentucky-5	Suspicious Activity	Jefferson	53172643	62314592

5 rows, 6 columns

Deliverables (Remaining Work)

- **Required Tasks (must be completed):**
 - Build SQLite database from csv files and load tables;
 - Write and test 3 python functions to:
 - parse outage event types [Complete]
 - parse FEMA disaster ids into standard identifiers, and

- normalize data across 3 ways of measuring impact (number of events, duration, customer-hours)
- Produce 3 visualizations: Scatterplot of outages vs income and gdp, a histogram to bin income categories vs relative frequency of outage, and heatmap to investigate correlations in outage and gdp datasets
- Complete GitHub repository with data, project notebook, revised README.md, and requirements.txt
- **Stretch Goals (optional, if time allows):**
 - Add a population dataset to generate ‘per capita’ income statistics
 - Create maps from outage and economic data

Project Timeline

- **Phase 1:**
 - Identify relevant datasets and develop database schema
 - Implement disaster id function, and perform data processing needed to meet schema
 - Implement database in SQLite3
- **Phase 2:**
 - Prepare 3 advanced SQL queries for exploratory analysis
 - Write python function to normalize outage impacts, to facilitate visualizations

- Draft scatterplot visualizations at county, state, and federal levels
- **Phase 3:**
 - Make and refine analytical conclusions, and additional visualizations
 - Add color and descriptions to improve visualizations
 - Document reproducible workflow in draft report
- **Phase 4:**
 - Review and polish repository structure and documentation
 - Refine markdown and narrative elements of project notebook
 - Submit Capstone Project GitHub repository

Additional Considerations

If time permits, population data will be sought to facilitate generation of ‘per capita’ economic statistics (instead of summary).